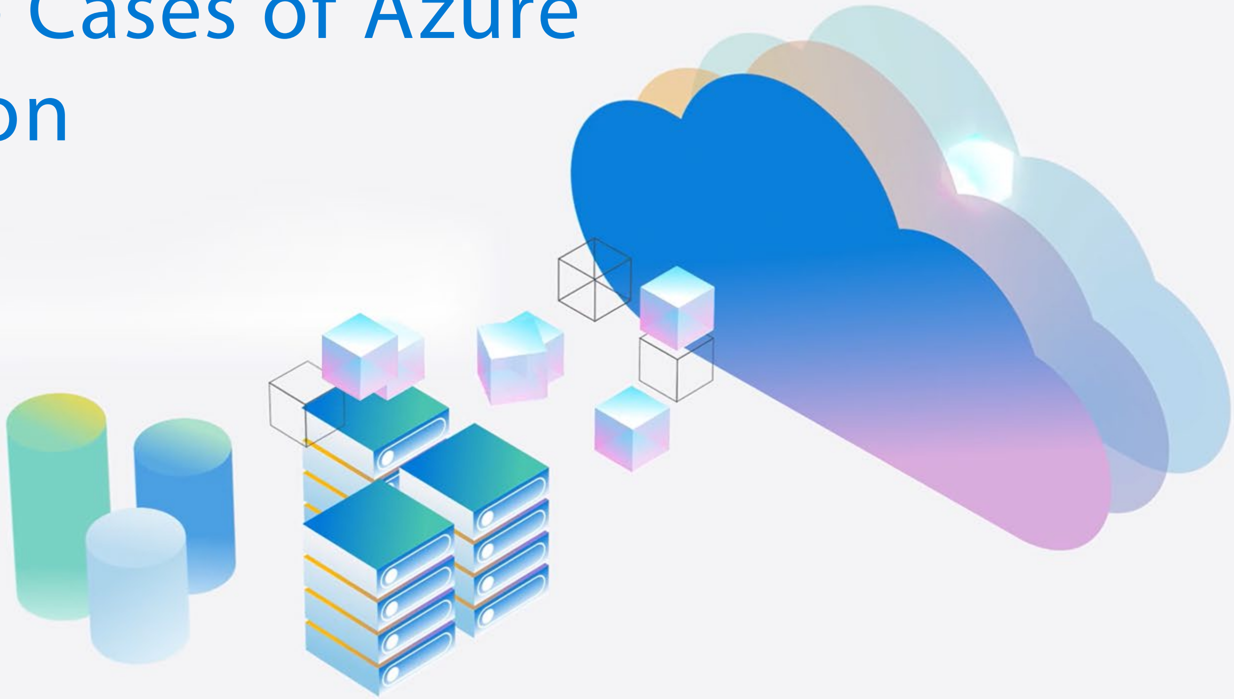
Real-World Use Cases of Azure VMware Solution

Why Choose AVS?
Exploring Use Cases, Licensing, SKU Pricing, and Overall Cost Assessment



Ruth Magyar
Kevin Jellow
Conner Loving

Agenda

- Why Organizations move to Cloud
- Considerations for AVS
- What workloads AVS can Target in FED
- How to get started – exploring a datacenter evacuation
- Tools to Help
- AVS TCO Deep Dive
- AVS Example Pricing
- Support – AVS VBD
- Recap

Reasons organizations move to cloud

Application modernization

Proximity to native Azure Services

Hybrid applications

Kubernetes standardization

Cloud migrations

Infrastructure refresh

Datacenter rightsizing, consolidation, retirement

Extended support updates/ hybrid benefits

Datacenter extension

Footprint expansion/ On-demand capacity

Unplanned projects

Extended support updates/ hybrid benefits

Cloud virtual desktops

Desktop bursting

Disaster protection

Proximity to cloud apps in Azure

Disaster recovery

New disaster recovery (DR) environment

Complement existing DR

Reduce/optimize DR cost

Meet with AVS team

Considerations for choosing AVS

Need for speed

Quick and secure migration

Ability to re-use VMware expertise & training

Cost optimization

Reduced CapEx expenditure

Cost optimization options with existing licenses

Innovation

Greater scale by modifying or extending apps

Operational consistency

Cloud competency

Meet with AVS team

What AVS can address in the DOD/FED Civ

Vsphere, vSAN, NSX Workloads

Migrate “As Is”

Unclassified only. Not available for SIPR/IL6 and above today

Connected, Datacenter Deployments

Not the right solution for disconnected workloads

Ex: Vehicle mounted, TSI, Weapon Systems

Virtual Desktops

Migrate Horizon deployments to Azure Virtual Desktop

Run Horizon on Azure

Run Horizon on Azure VMware Solution

Meet with AVS team

Getting Started: Exploring a Datacenter Evacuation

Current Environment Analysis

VMware currently deployed – vSphere, Horizon, VSAN etc

Exit Timeline

Workload Operating Systems & Versions

Extended Security Updates

Azure Hybrid Use Benefit

Meet with AVS team

AVS on your Cloud Contract

- AVS Pricing locked in for 5 years
- VMWare licenses included in AVS cost, additional VMWare licenses not required
- Cloud Contract Discount may apply (JWCC, CAMO, etc)
- No need for additional hardware refresh
- Only pay for the compute you need
- Free Extended Security Updates
- Bring your Windows and SQL licenses for additional savings

Meet with AVS team

Tools to Help

RVTools Export

What is it and what does it provide

Why is it more effective vs Cores/RAM/Storage (TB)

Enables us to most effectively “pack the dishwasher”

Meet with AVS team

TCO Factors to Consider

On Prem

- Annual on-prem license cost
- Hardware refresh costs
- Extended Security Updates
- SQL/WIN Server License costs
- Labor costs
- Power/cooling/etc (not always a factor in GOV)
- Datacenter physical security

In AVS

- AVS Nodes
- SQL/WIN Server License costs
- Cloud admin/labor

Meet with AVS team

Contoso SOFTWARE INC - AV64's VMware Overview

This view provides a snapshot of the VMware infrastructure scanned via RVTools and will form the basis of the overall assessment.

Note: Mainframe, AIX, Solaris, and NAS storage are out of scope of this assessment.

ESX HOSTS

171

vCenters Scanned

- HE1-Datacenter
- HE2-Datacenter
- VxRail-Datacenter

VIRTUAL MACHINES

5818

Window	1411
RHEL	3932
Linux & Other	475

UTILIZATION

26,306
Allocated Cores
100% Utilization

211,515
Allocated RAM (GB)
100% Utilization

3,887
Used Storage (TB)

SUPPORT STATUS

Out of Support

Linux	59
Centos	147
Windows	77

MIGRATION CALLOUTS

Readiness

- READY 5566
- CONDITIONS 224
- BLOCKED 28

Power Status

Executive Summary - AVS TCO Savings

Analysis Period	Servers	Cost Savings	Cost Reduction	Effort Saved	CO_2 Reduction
3 Years	5818	$11.0M	29%	29 Months	97%

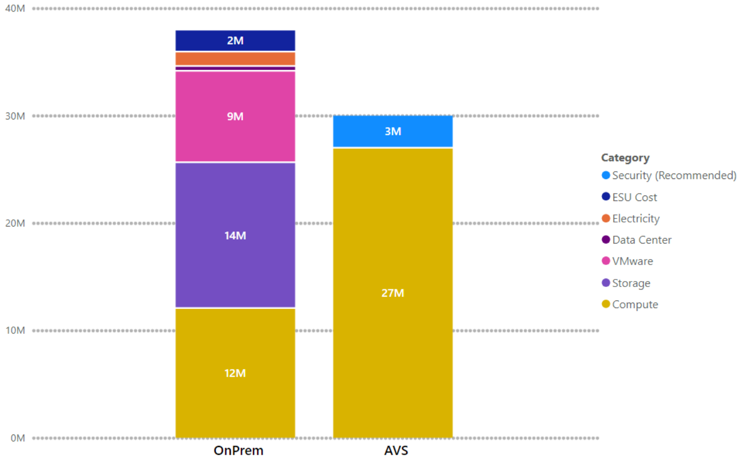
Total Cost of Ownership

Category	On-Premises	AVS
Compute	$11,995,808	$26,925,859
Storage	$13,575,647	$0
VMware	$8,518,650	$0
Data Center	$445,200	$0
Electricity	$1,340,202	$0
ESU Cost	$2,046,530	$0
Total	$37,922,036	$26,925,859
Security (Recommended)		$3,057,941

Costing Assumptions

See Assumptions Slide at end for more information
No powered off servers included
All servers are assumed to be Production
Disaster Recovery is out of scope for this view
Backup is out of scope for this view

Cost Transformation



[Azure TCO Calculator]

Scenario 1
Azure VMware Solution

205

Total AV64 Nodes

3 Year Estimate

On Premise Cost	Cloud Hosting Cost
$37.9M	$26.9M

3 Year Savings
$11.0M

Key Benefits

- Ease of Migration
- Speed
- Cost Savings
- Simplification

Financial Overview - Azure VMware Solution

The following table provides a breakdown of an AVS hosted solution, applying assumptions from Assumptions appendix slide.

On-Prem Server Workload	Azure Recommendation	Device Count	Compute Cost, Monthly	Storage Cost, Monthly	Total Cost, Monthly	Comments
VMware-hosted Virtual Machines	Azure VMware Solution	5818 Servers	$747,941	$0	$747,941	205x AV64 AVS Nodes (64 cores, 1024 GB RAM, 21.12 TB All Flash Storage, each) - 3 Year Reserved Instance & 3x AV36P Nodes (36 Cores, 768 GB RAM, 19.2 TB All Flash Storage, each) with 25% AV64 field empowerment discount applied
Total			**$747,941**	**$0**	**$747,941**	

Optional Service	Azure Recommendation	Device Count	Base Cost, Monthly	Storage Cost, Monthly	Total Cost, Monthly	Comments
Server Security	Microsoft Defender for Cloud Servers P2	5818 Servers	$84,943	$0	$84,943	Utilize Defender for Cloud to protect 5818 servers at a cost of $15/Month per Server

*Optional Services have not been added to the total Cloud Hosting Cost at this stage.

Scenario 2

Rehost All on IaaS

5818 Servers | 3.9 PB

3 Year Estimate

On Premise Cost	Cloud Hosting Cost
$37.9M	$36.3M

3 Year Savings

$1.6M

Key Operational Benefits

- Cost Savings
- Flexibility
- Security
- Redundancy
- Operational Savings
- Availability

Financial Overview - Rehost All on IaaS

The following table provides a breakdown if all servers were to be right-sized and Rehosted in Azure.

On-Prem Server Workload	Azure Recommendation	Device Count	Compute Cost, Monthly	Storage Cost, Monthly	Total Cost, Monthly	Comments
Production Servers	Azure VM	5818 Servers	$772,511	$235,190	$1,007,700	AHB + 3 Year Reserved Instance
Total			**$772,511**	**$235,190**	**$1,007,700**	

Optional Service	Azure Recommendation	Device Count	Base Cost, Monthly	Storage Cost, Monthly	Total Cost, Monthly	Comments
Server Security	Microsoft Defender for Cloud Servers P2	5818 Servers	$84,943	$0	$84,943	Utilize Defender for Cloud to protect 5818 servers at a cost of $15/Month per Server

*Optional Services have not been added to the total Cloud Hosting Cost at this stage.

The cost of attached disks appear under Storage cost column.

[Azure Hybrid Benefit] has been applied to all production servers allowing you to save up to 40% on Windows, SQL and RHEL VMs.

Additional savings can be gained by applying Dev/Test licensing to non-production servers.

Migration Acceleration Simulation

Azure VMware Services (AVS) allows customers to quicky and efficiently move out of On-Premises VMware.

Action to Save

Migrate to Azure VMware Services (AVS) rather than any cloud IaaS solution. This will greatly reduce the number of months needed to migrate from On-Premises VMware and mitigate Broadcom price changes.

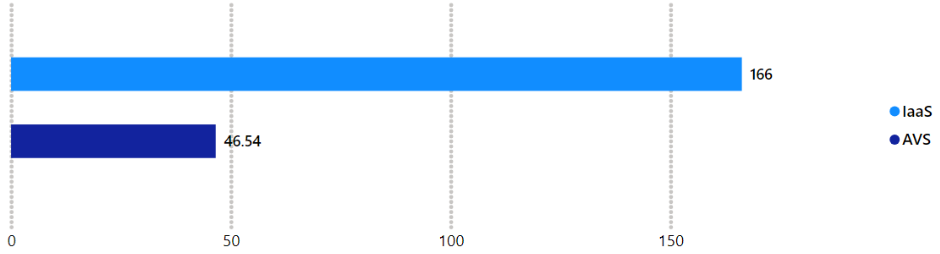
$985.3K

ROM additional FTE Savings

Compared to IaaS Migration

The following depicts a migration scenario comparison, highlighting the simplicity and speed of an AVS based migration.
Assumed migration velocity: 125 AVS servers migrated a week vs 35 IaaS servers.
Assumed migration team size is 6 FTE at a salary of 70,000 USD per FTE per year.

Migration Duration (Weeks) IaaS vs Azure VMware Solution (AVS)



Migration Simulation	AVS	IaaS
Time to Complete Rehosting	11.64 months	41.56 months
Estimated Completion Date	18 Aug 2025	04 Dec 2027

Potential Migration Squad Resource Savings	
ROM Resource Saving	13.8 FTE
ROM Resource Cost saving	$985.3K
ROM Time Saving	29.92 Months

Comparison only includes the migration of servers and not the design / preparation time as these are expected to be the same regardless of approach.

Azure VMWare Solution (AVS) | Building blocks

Enhanced Solutions

These services can be delivered on the customer's Unified base contract, through Azure IaaS EDE, or as part of the Migrate VMware to AVS Designated Engineering (DE) offering.

1 Azure Landing Zone (ALZ)

Build a secure and scalable architecture aligned to the Cloud Adoption Framework across eight critical design areas.

WorkshopPLUS: Azure Landing Zone
Learn about the ALZ design principles and reference architecture and options for implementation.

Azure Landing Zone Assessment
Prepare for your ALZ implementation by assessing the key decision points that will influence your Azure architectural design.

Azure Landing Zone Deployment
Implement a landing zone from ALZ reference architecture to scale as your Azure workload needs evolve.

2 Migrate and Modernize VMware to AVS

Migrate or extend your on-premises VMware Solution to Azure and modernize your applications with Azure native services.

Plan & Design

Workshop – Azure VMWare Solution
Learn about the best practices for architecture and configuration in your AVS environment for a successful implementation.

Architecture Design and Review session
Design or assess the technical architecture for the AVS environment with careful examination for Enterprise Scale's critical design areas.

Proof of Concept
Test a functional deployment of a single AVS private cloud that integrates with your existing infrastructure (Network, Active Directory Domain Services, Azure Monitor).

Deploy

Technical Blocker Mitigation
Address existing blockers on AVS adoption and migration.

Production Deployment Oversight
Ensure the AVS solution architecture deployed to production is validated and supported.

Go-Live Assessment
Assess your pre-production AVS workload to create a go-live plan and prepare for a successful deployment.

3 Optimize

Optimize your existing AVS workload to ensure a resilient architecture that maximizes workload stability.

Review and Remediate

Well-Architected Reliability Assessment
Perform an in-depth assessment of your workload to validate reliability based on best practices from the Well-Architected Framework. Identify gaps in architecture and resources for critical resiliency, networking, availability, and recovery risks.

Well Architected Review and Remediation for Azure VMWare Solution
Execute a comprehensive review and develop a remediation plan for your Azure VMWare Solution, aligned to the five pillars of Well-Architected Framework.

Refer to Health and Optimization for additional service recommendations and Well-Architected Framework pillar specific Pathways.

AVS | Migration and Modernization Pathway View

Enhanced Solutions

These services can be delivered on the customer's Unified base contract, through the Azure IaaS EDE package, or as part of the Migrate VMware to AVS Designated Engineering (DE) offering.

Customers can enter this pathway at different stages based on their technical needs.

Our organization is looking to migrate or extend our on-premises VMware Solution to Azure.

My team needs to learn about AVS features and operations.

WorkshopPLUS: Azure VMWare Solution

I want to implement an Azure Landing Zone to ensure governance and scalability.

Azure Landing Zone Deployment

I need to validate the feasibility of the AVS solution, or I need to test an Azure product in AVS.

Proof of Concept for Migrating and Modernizing VMware Solution

I want Microsoft to help design or review my AVS solution.

Architecture Design and Review for Migrating and Modernizing VMware Solution

I am deploying AVS and finding some blockers that prevent me from completing implementation.

Technical Blocker Mitigation for Migrating and Modernizing VMware Solution

I am deploying AVS to production and looking for Microsoft to validate that I am following best practices.

Production Deployment Oversight for Migrating and Modernizing VMware Solution

The CSA should use Value Areas if delivering these activities on Enhanced Designated Engineering (EDE).

I am looking for Microsoft to evaluate whether the AVS deployment adheres to best practice standards for production readiness.

Go-Live for Migrating and Modernizing VMware Solution

I want to ensure my AVS implementation is reliable.

Well-Architected Reliability Assessment

I want to assess and remediate my existing AVS workload across the five pillars of the Well-Architected Framework.

Well-Architected Review and Remediation for Azure VMware Solution

I want to further optimize and implement recommendations for my AVS workload.

Health and Optimization Pathway

Optimization cycle

Recap

1. Reduce Costs

2. Migrate with velocity

3. Hybrid enable your workloads

Meet with AVS team